UMER SIDDIQUI

Backend & AI Systems Engineer | System Design & Agentic AI

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PROFESSIONAL SUMMARY

Backend and AI Systems Engineer with production experience building scalable, low-latency LLM-powered applications — agentic workflows, RAG pipelines, and enterprise voice AI — from architecture through cloud deployment. Strong grounding in core system design (API architecture, database selection, caching, load balancing, authentication/security) combined with applied AI infrastructure experience (model serving, orchestration, LLMOps, cost and latency trade-offs). Comfortable owning systems end-to-end: designing the architecture, writing the backend, and reasoning about how it behaves at scale. Strong Data Structures & Algorithms fundamentals, applied through consistent problem-solving practice.

Core Stack: Python · FastAPI · Node.js/NestJS · TypeScript · PostgreSQL · pgvector · Redis · OAuth 2.0/JWT · Docker · AWS · GCP · CI/CD · LangGraph · n8n · Anthropic Claude · RAG · System Design · DSA

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Saynt AI **06/2025 – Present · Remote**

– Owned end-to-end backend development for the voice agent dashboard and a dental practice management (PMS) platform — REST APIs, business logic, and authentication built in NestJS/TypeScript, deployed on AWS.

– Implemented authentication from the ground up, including Google OAuth login, across NestJS and FastAPI services.

– Set up the CI/CD pipeline for the NestJS backend using GitHub Actions — merges to main auto-deploy to AWS, removing manual release steps.

– Designed agentic workflows with LangGraph and RAG pipelines with webhook-based tool-calling, integrating Twilio telephony and PMS backends (OpenDental, Dentrix) for low-latency, context-aware voice interactions.

Jr. AI Developer | Erly Stage Studios **10/2024 – 06/2025 · Remote**

– Built the full backend for the WhatsApp Meeting Scheduling Agent in FastAPI — webhook handling, agent orchestration (LangGraph + OpenAI), and state management.

– Deployed the service on GCP with Firestore as the primary data store; collaborated with an Anthropic engineer on system architecture and scalability decisions.

– Built a product Q&A chatbot API in Flask over scraped product data in PostgreSQL, combining direct SQL queries with pgvector similarity search for hybrid structured + semantic retrieval; handed off the API for frontend integration.

AI & Data Analytics Intern | NCL NEDUET **04/2024 – 10/2024 · Karachi**

– Delivered AI model development, EDA, and ML optimization projects with real-world business applications.

– Built LocoBot, a lab assistant robot combining real-time face recognition with a RAG chatbot for booking meetings with faculty; used directly by students visiting the lab.

KEY PROJECTS

Autonomous WhatsApp Meeting Scheduling Agent

FastAPI · LangGraph · OpenAI · GCP Firestore · Webhooks · State Management

Production event-driven conversational agent orchestrating multi-step calendar booking via webhooks, state persistence, and dynamic tool calling with real-time error recovery.

Autonomous Icon CAPTCHA Verification Engine

YOLO (Fine-Tuned) · ResNet (Fine-Tuned) · PyTorch · OpenCV · Spatial Geometry

End-to-end computer vision pipeline solving multi-stage icon verification. Fine-tuned YOLO for target symbol detection and ResNet for orientation alignment, computing click coordinates with >98% accuracy.

Enterprise Voice AI Telephony Agent

ElevenLabs · Twilio · LangChain · RAG · Practice Management Systems

Intelligent appointment scheduler with RAG context retrieval, webhook tool-calling, and real-time bidirectional telephony integration, designed for low-latency production use.

Final Year Project — Novel Actor-Critic Algorithm for Reinforcement Learning

Reinforcement Learning · Actor-Critic · PyTorch · OpenAI Gym

Proposed a modification to the critic function in standard Actor-Critic architectures, achieving state-of-the-art performance surpassing prior baselines across multiple OpenAI Gym environments.

EDUCATION

B.E. Electrical Engineering — NED University of Engineering & Technology · CGPA: 2.6/4.0 · 02/2022 – 08/2026

Higher Secondary (Intermediate) — Bahria College Karsaz · BIEK Board · 996/1100 marks · 2020 – 2022

Secondary School (Matriculation) — CAA Model School and College, Karachi · BSEK Board · 755/850 marks · 2018 – 2020